

Michael Khavkin

Ph.D. Candidate

School of Industrial & Intelligent Systems Engineering, Faculty of Engineering,
Tel Aviv University
Israel

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Research Interests

- Usable Security and Privacy
- Differential privacy
- Human-computer Interaction (HCI)
- Machine learning

Education

- **Tel Aviv University** Tel Aviv, Israel
Ph.D. Candidate, Department of Industrial Engineering, Faculty of Engineering
– Thesis title: User-oriented Differential Privacy
– Advisor: Prof. Eran Toch
2022 – Present
- **Ben-Gurion University of the Negev** Be'er-Sheva, Israel
M.Sc. (summa cum laude); Software and Information Systems Engineering
– Thesis title: Differential Privacy-Preserving Data Stream Publishing
– Advisor: Prof. Mark Last
2017–2018
- **Ben-Gurion University of the Negev** Be'er-Sheva, Israel
B.Sc. (magna cum laude); Software and Information Systems Engineering
2013–2017

Awards and Honors

1. **Travel Grant**, *The 6th Annual Symposium on Applications of Contextual Integrity*, Rutgers University, 2024.
2. **Teaching Excellence Award**, *Department of Industrial Engineering, Tel Aviv University*, 2023.
3. **Westin Scholar Award for excellence in Privacy**, *The International Association of Privacy Professionals (IAPP)*, 2022.
4. **Dean's Excellence Award, inclusion in the Dean's List**, *B.Sc., Department of Software and Information Systems Engineering, Ben-Gurion University of the Negev*, 2017.

Journal Publications

1. **Khavkin, M.** & Toch, E. (2025). Differential Privacy Configurations in the Real World: A Comparative Analysis. *IEEE Transactions on Knowledge and Data Engineering*, 37(11), 6334–6351, IEEE.
<https://doi.org/10.1109/TKDE.2025.3603731>
2. **Khavkin, M.** & Toch, E. (2025). Investigating the impact of differential privacy obfuscation on users' data disclosure decisions. *Decision Support Systems*, 196(C), 114474, Elsevier, North Holland, Netherlands
<https://doi.org/10.1016/j.dss.2025.114474>
3. Cohen, M., **Khavkin, M.**, Movsowitz Davidow, D., & Toch, E. (2024). ChatGPT in the Public Eye: Ethical Principles and Generative Concerns in Social Media Discussions. *New Media & Society*, 28(1), 5–31, SAGE Publications. London, United Kingdom. <https://doi.org/10.1177/14614448241279034>

Conference Publications

1. **Khavkin, M.**, & Toch, E. (2023). Valuation of Differential Privacy Budget in Data Trade: A Conjoint Analysis. In *Privacy and Identity Management: 17th IFIP WG 9.2, 9.6/11.7, 11.6/SIG 9.2. 2 International Summer School, Privacy and Identity 2022, Virtual Event, Proceedings*, 69–84, Springer Nature, Cham, Switzerland.
https://doi.org/10.1007/978-3-031-31971-6_7

2. **Khavkin, M.**, & Last, M. (2018). Preserving Differential Privacy and Utility of Non-Stationary Data Streams. In *2018 IEEE International Conference on Data Mining Workshops (ICDMW)*, 29–34, IEEE, Singapore.
<https://doi.org/10.1109/ICDMW.2018.00012>

Chapters in Books

1. Brodt, O., **Khavkin, M.**, Rokach, L., Shabtai, A., & Elovici, Y. (2021). Artificial Intelligence and (the Lack of) Security. In *Leading Legal Disruption: Artificial Intelligence and a Toolkit for Lawyers and the Law*, Thomson Reuters, Toronto, Canada.

Patents

- Elovici, Y., Shabtai, A., Brodt, O., Mimran, D., **Khavkin, M.**, & Grolman, E. (2021). A Method and a System for Testing Machine Learning and Deep Learning Models for Robustness, and Durability Against Adversarial Bias and Privacy Attacks. *European Patent Office*. EU Patent No. EP3910479A1, Munich, Germany.
- Elovici, Y., Brodt, O., Shabtai, A., Grolman, E., Mimran, D., & **Khavkin, M.** (2024). A Method for Assessment of the Robustness and Resilience of Machine Learning Models to Model Extraction Attacks on AI-based Systems. *European Patent Office*. EU Patent No. EP4365787A1, Munich, Germany.

Participation in Conferences, Workshops, Posters and Abstracts

1. **Michael Khavkin**, Eran Toch. Research paper: “Investigating the impact of differential privacy obfuscation on users’ data disclosure decisions”, *The 12th Israeli Human-Computer Interaction Research Conference (IsraHCI2025)*, Haifa University, Haifa, Israel, October 2025.
2. **Michael Khavkin**, Eran Toch. Research paper: “Investigating How User Disclosure Decisions Shape Machine Learning Performance Under Differential Privacy”, *Israel Association for Information Systems (ILAIS)*, Ben-Gurion University, Be’er Sheva, Israel, July 2025.
3. **Michael Khavkin**, Eran Toch. Research paper: “The Impact of Privacy Guarantees and User Decisions on the Performance of Machine Learning Models”, *The 24th National Conference on Industrial Engineering and Management*, Open University, Ra’anana, Israel, April 2025.
4. **Michael Khavkin**, Eran Toch. Research paper: “Investigating the Impact of Differential Privacy on Users’ Contextual Data Disclosure Decisions”, *The 6th Annual Symposium on Applications of Contextual Integrity*, Rutgers University, New Jersey, US, September 2024.
5. **Michael Khavkin**, Eran Toch. WIP abstract: “Valuation of Differential Privacy in Data Trade: A User Study”, *The 10th Israeli Privacy Conference*, Tel Aviv University, The Edmond J. Safra Center for Ethics, Tel Aviv University, Tel Aviv, Israel, May 2023.
6. **Michael Khavkin**, Eran Toch. Research paper: “Trade-Offs in Data Sharing Under Personalized Differential Privacy: A Conjoint Analysis”, *The 17th IFIP Summer School on Privacy and Identity Management*, Virtual, September 2022.
7. **Michael Khavkin**, Mark Last. Research paper: “Preserving Differential Privacy and Utility of Non-Stationary Data Streams”, *2018 IEEE International Conference on Data Mining Workshops (ICDMW)*, Singapore, November 2018.
8. **Michael Khavkin**, Mark Last. Extended abstract: “Preserving Differential Privacy and Utility of Non-Stationary Data Streams”, *The 2nd International Symposium on Cyber Security Cryptography and Machine Learning (CSCML 2018)*, Be’er Sheva, Israel, June 2018.
9. **Michael Khavkin**, Mark Last. Poster: “Preserving Differential Privacy and Utility of Non-Stationary Data Streams”. *The 6th Conference on Data Mining for Business Intelligence*, Be’er Sheva, Israel, May 2018.

Professional Experience

- **Explorium** Tel Aviv, Israel
Data scientist 2021–2022

- Development of machine learning models for classification and regression prediction, leveraging external data sources to boost performance, for various use cases, including insurance claim prediction and loan approval

- **Telekom Innovation Laboratories (Security Research Center @ BGU)** Be'er-Sheva, Israel
Data scientist and research Assistant 2019 – 2021
 - Project topic: Testing machine learning and deep learning models for robustness, and durability against adversarial bias and privacy attacks
 - External collaboration with Deutsche Telekom and T-Mobile Austria
 - Designing, implementing and evaluating a testbed for detection of adversarial bias and privacy attacks

Teaching Experience

- **Lecturer** *Tel Aviv University*
 - *Undergraduate courses*
 - Design of Data Base Systems – 2026
- **Teaching Assistant** *Ben-Gurion University of the Negev; Tel Aviv University*
 - *Undergraduate courses*
 - Data Mining and Data Warehousing – 2017–2018
 - Algorithms – 2017–2018
 - Analysis and Design of Information Systems – 2022
 - Design of Data Base Systems – 2023
 - Big Data Engineering – 2023–2026
 - Information Retrieval and Recommender Systems – 2024–2026
 - Statistical Foundations of Data Science – 2024
 - Data Ethics – 2024
- *Graduate courses* *Tel Aviv University*
 - Big Data Technologies and Databases – 2023–2026

Service

- *Chair assistant*
 - Subcommittee (SC) assistant: “Privacy and Security” subcommittee for the *ACM CHI Conference on Human Factors in Computing Systems, 2025, 2026* (CHI’2025, CHI’2026)
- *Reviewer*
 - *International Conference on Machine Learning (ICML), 2024* (1 paper).
 - *ACM CHI Conference on Human Factors in Computing Systems, 2023* (2 papers).
 - *Journal of Information Security and Applications (JISA), 2018* (1 paper)
 - *IEEE Transactions on Cybernetics Journal, 2017* (1 paper)

References

- **Prof. Eran Toch** erant@tauex.tau.ac.il
Associate Professor, School of Industrial & Intelligent Systems Engineering, Tel Aviv University